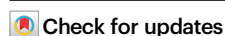


# Enantioselective palladium-catalyzed carbonylation of gem-dibromoalkenes to chiral benzo[7]annulenes

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Chemo- and enantioselective mono-debrominative functionalization of 2,2-disubstituted 1,1-dibromoalkenes to access tetrasubstituted vinyl bromides remains a formidable challenge, owing to competitive double functionalization and the difficulty in achieving enantiocontrol when the *gem*-dibromoalkenyl moiety embodies a prochiral plane. Herein, we report a palladium-catalyzed asymmetric monocarbonylation of conformationally rigid, prochiral benzo[7]annulene-based *gem*-dibromoalkenes, furnishing inherently chiral tetrasubstituted benzo[7]annulene derivatives with excellent yields (up to 97%) and high enantioselectivities (up to 98:2 er). Notably, the inclusion of simple acid additives enables a switchable pathway divergence to double debrominative carbonylation, delivering inherently chiral trisubstituted vinyl carboxylic acids with equally good chemo- and enantiocontrol. This protocol is also applicable to unsymmetric *gem*-dibromoalkene substrates, affording the corresponding products with synthetically useful yields and enantioselectivities. Mechanistic studies, supported by DFT calculations, reveal that the acid-induced formation of a key alkylidene carbene–Pd–H species is pivotal for the pathway switch, followed by enantioselective carbene insertion into the Pd–H bond and subsequent CO insertion. This work not only provides a modular and divergent route to both tri- and tetrasubstituted inherently chiral benzo[7]annulene scaffolds, but also represents the first asymmetric desymmetrization of planar prochiral *gem*-dibromoalkenes via palladium catalysis.

2-Substituted and 2,2-disubstituted 1,1-dibromoalkenes, readily available from the corresponding aldehydes and ketones via Wittig-type olefination<sup>1</sup>, are well recognized as versatile building blocks in the synthesis of alkynes, polysubstituted alkenes, heterocycles, and natural products<sup>2–4</sup>. The presence of two geminal C(sp<sup>2</sup>)-Br bonds renders them reactive substrates in transition-metal-catalyzed cross-coupling reactions; however, chemoselectivity between mono- and double-

debrominative functionalization arises. In the case of aldehyde-derived *gem*-dibromoalkenes, activation of the *E*-bromide is more sterically and electronically favorable, as demonstrated in Pd-catalyzed monodebrominative hydrogenation, Suzuki coupling, carbonylation, etc.<sup>3</sup>. On the other hand, chemoselective transformation of one of the vinyl bromides in 2,2-disubstituted 1,1-dibromoalkenes, giving *tetra*-substituted bromoalkenes, remains a challenging task (Fig. 1A)<sup>5–7</sup>. It is

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believed that there are two inherent challenges in this regard: 1) Competitive Fritsch–Buttenberg–Wiechell (FBW) rearrangement leads to thermodynamically more stable alkynes via vinylidene carbene intermediates under basic or transition-metal conditions accompanied by side reactions such as carbene dimerization<sup>8–15</sup>; and 2) Double-functionalization is always unavoidable, even when the stoichiometry of reagents is strictly controlled<sup>16–18</sup>.

Over the past decades, catalytic asymmetric desymmetrization of prochiral substrates has made significant progress to construct centrally, planarly, and axially chiral molecules (Fig. 1B)<sup>19–22</sup>. In this process, two identical functional groups can be differentiated chemically by forming diastereomeric intermediates with the chiral catalysts. This strategy was also successfully applied to synthesize inherently chiral calix[4]arenes by desymmetrizing C–H functionalization<sup>23,24</sup>. However, with growing interest in inherently chiral medium-sized rings, methods for their enantioselective preparation have remained underexplored<sup>25–33</sup>. Given the unique attributes of inherently chiral medium-sized molecules in chiral recognition, asymmetric catalysis, and materials science, developing new synthetic strategies, such as catalytic asymmetric desymmetrization, has become an urgent and pivotal challenge in the field<sup>34–37</sup>.

In axially chiral allenes and alkylidene compounds, the absolute configuration of the axial chirality reverses by exchanging the positions of the substituents on the germinal alkylidene carbon. In other words, these molecules with two identical groups at the alkylidene carbon are prochiral. It is hypothesized that a prochiral medium-sized ring with a rigid conformation equipped with *gem*-dibromoalkylidene moiety could be designed, and differentiation of the two C(sp<sup>2</sup>)-Br bonds could then be achieved under palladium catalysis in the presence of a chiral ligand (Fig. 1C). An additional benefit of this strategy is that the remaining C(sp<sup>2</sup>)-Br bond provides potential for diversifying the scaffold.

Continuing our interest in the synthesis of inherently chiral molecules<sup>38–40</sup> and palladium-catalyzed selective carbonylation reactions, we envisioned that chemo- and enantioselective monocarbonylation of *gem*-dibromoalkenes on a conformationally inflexible scaffold could open new avenues to inherently chiral molecules. Herein, we report a palladium-catalyzed divergent asymmetric carbonylation of benzo[7]annulene-based *gem*-dibromides (Fig. 1D). The remaining vinyl bromide in the products can be diversified further to give tetrasubstituted alkenes, which may serve as inherently chiral ligands or catalysts.

## Results

### Reaction optimization

To verify the hypothesis, a preliminary screening of nucleophiles was first conducted in the Pd-catalyzed carbonylation reaction of benzo[7]annulene-derived *gem*-dibromide **a-1** (See Supplementary Table S1 for details). When phenols were employed as nucleophiles, the desired carbonylative product was obtained as the major product in low yield, accompanied by products from double carbonylation and debrominative monocarbonylation. With the preliminary results in hand, chiral bisphosphine ligands were systematically evaluated using *p*-cresol (**b-1**) as the nucleophile (See Supplementary Table S2 for details). The results indicated that most of the bisphosphine ligands screened promoted the formation of the desired product; however, yields were generally low, with moderate to poor enantioselectivities. Gratifyingly, the MeO-biphep ligand **L2**, bearing 2-furyl substituents, afforded the mono-carbonylative product **c-1** chemoselectively in low yield (25%), but with excellent enantioselectivity (96:4 er). Interestingly, upon the addition of one equivalent of PivOH under otherwise identical conditions, the debrominated product **d-1** was obtained as the major product in moderate yield and enantioselectivity (45%, 81.5:18.5 er) (See Supplementary Table S5 for details).

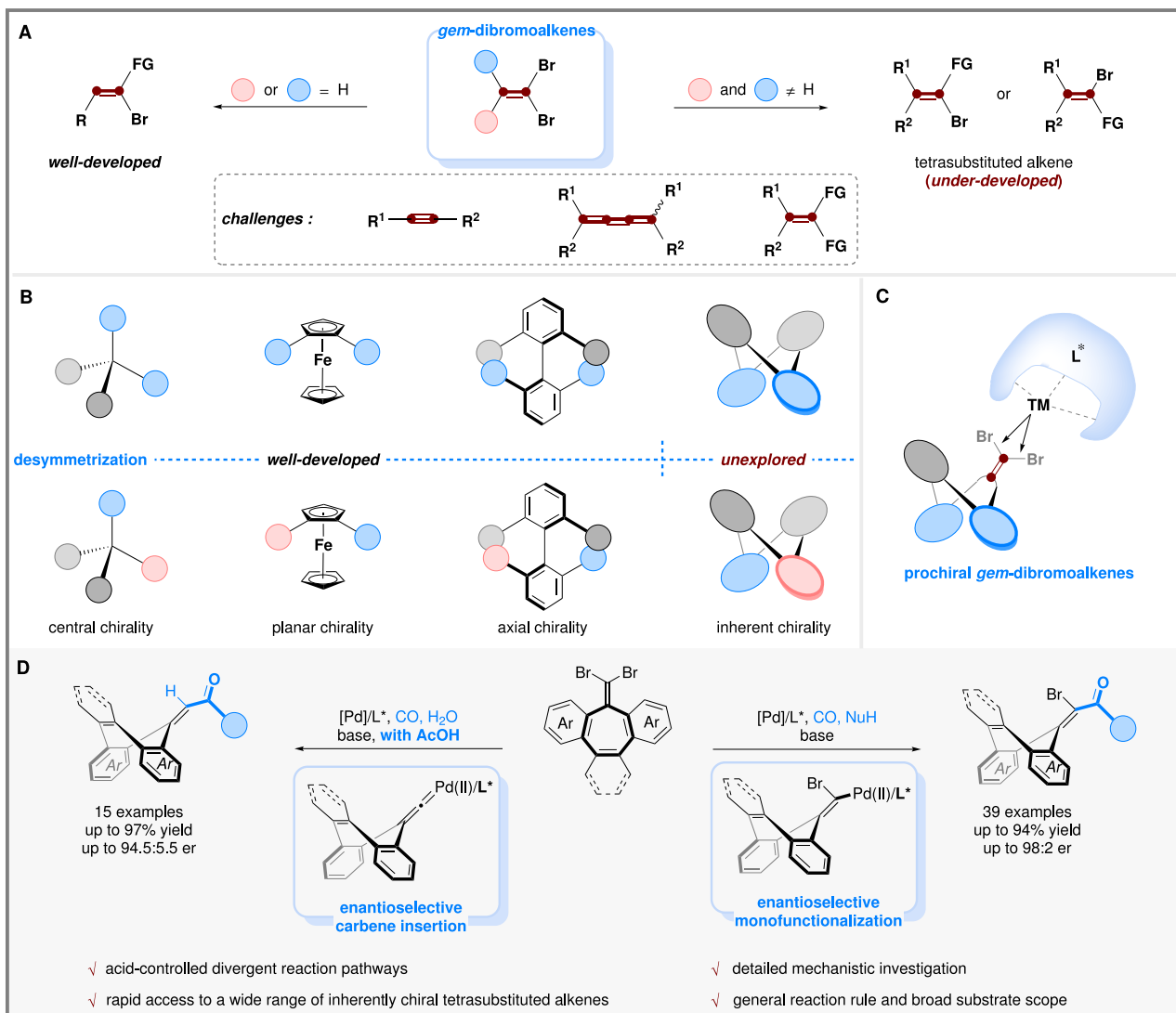
Following the identification of the optimal chiral ligand **L2**, solvents were screened, resulting in comparable enantioselectivity but significantly diminished yields (entries 1–3). Subsequent evaluation of bases revealed that only strong inorganic bases, such as NaH, KOH, and Cs<sub>2</sub>CO<sub>3</sub>, facilitated the formation of **c-1** (entries 4–6). Reactions at elevated temperatures led to a decreased yield of **c-1** despite complete consumption of the starting material, which was attributed to hydrolysis of the phenol ester to the corresponding carboxylic acid (entry 7). Screening of palladium sources with Cs<sub>2</sub>CO<sub>3</sub> as the base revealed that Pd(OAc)<sub>2</sub> emerged as the catalyst of choice (entries 8–11). Ultimately, the optimal conditions for the formation of **c-1** (73% yield, 96.5:3.5 er) were identified by adding molecular sieves to mitigate product hydrolysis and increasing the equivalents of both the nucleophile and the base (entry 12). It is intriguing that the debrominated by-product **d-1** is almost racemic under all of the reaction conditions during optimization (entries 1–12).

Given the structural novelty of chiral carboxylic acid **d-2**, a detailed screening of acidic additives aiming at chemo- and enantioselective formation of **d-2** was conducted (See Supplementary Table S5 to S6 for details). The combination of K<sub>2</sub>CO<sub>3</sub> and acid (AcOH or PivOH) gave similar results with KO<sub>2</sub>Piv, whereas PivOH alone or rigorous dehydration conditions completely inhibited the reaction (**d-1** as the major product, entries 13–17). Subsequently, water was directly employed as the nucleophile in place of **b-1**, yielding the corresponding carboxylic acid **d-2** in good yield and enantioselectivity in THF as the solvent (entries 18 and 19). Finally, fine-tuning the water stoichiometry enabled highly efficient and chemoselective formation of the debrominated carboxylic acid **d-2** with excellent enantioselectivity (93% yield, 94.5:5.5 er; Table 1, entry 20).

### Substrate scope of the monocarbonylation reaction

With the optimal reaction conditions established, the scope of phenolic nucleophiles was then evaluated (Fig. 2). Phenol esters bearing substituents at the *para*- (**c-1** to **c-6**), *meta*- (**c-7** and **c-10**), or *ortho*-positions (**c-8**, **c-9**, and **c-11**) were obtained in 51%–80% yields with enantioselectivity (er) values ranging from 95:5 to 96.5:3.5. These results demonstrate excellent functional group tolerance of this protocol towards methyl, methoxy, halogen (F, Cl), acetyl, and naphthyl groups, as well as substrates containing highly reactive allyl units (**c-9**) or those with significant steric hindrance (**c-11**). Overall, substituents on the phenolic nucleophiles had minimal effect on the enantioselectivity of the reaction. Furthermore, this method was successfully applied to the modification of bioactive molecules containing phenolic hydroxyl groups, including tetraphenylethylene derivatives, estrone, and paracetamol, yielding products **c-12** to **c-14** in 54%–69% yields with excellent stereoselectivity. Interestingly, when aliphatic amines or electron-rich aromatic amines with moderate nucleophilicity were added to the standard reaction system together with 3-methoxyphenol, the resulting ester products underwent in situ aminolysis, delivering the corresponding amide products (**c-15** to **c-18**) in good yields and excellent enantioselectivity. The absolute configuration of compound **c-15** was unambiguously assigned via single-crystal X-ray diffraction analysis. The strategy of combining nucleophiles effectively compensates for the limitations associated with using amino nucleophiles alone.

Next, the substrate scope of benzo[7]annulene *gem*-dibromides was examined. For tribenzo[*a,c,e*][7]annulene systems bearing different substituents at the symmetric 10,13-positions, the reaction exhibited significant electronic dependence (**c-19** to **c-22**). Electron-withdrawing groups favored both reaction efficiency and enantioselectivity (**c-21**: 94% yield, 97:3 er) (See Supplementary Fig. S3–S4 for details). When the 7,11-positions were substituted with methyl groups, the reaction proceeded smoothly (**c-23**). However, introducing sterically demanding *p*-tolyl groups led to diminished efficiency (**c-24**). Substrates with fused aromatic or heteroaromatic rings opposite to



**Fig. 1 | Palladium-Catalyzed Asymmetric Transformations of *gem*-Dibromides.** **A** Selective Mono-Functionalization of *gem*-Dibromoalkenes. **B** Enantioselective Desymmetrization of Diverse Prochiral Compounds. **C** Chemoselective Differentiation of Conformationally Rigid *gem*-Dibromoalkenes Controlled by Chiral

Ligands. **D** This Work: Palladium-Catalyzed Asymmetric Carbonylative Transformation of Benzo[7]annulene-based *gem*-Dibromides: Divergent Reaction Pathways Controlled by Acidic Additives.

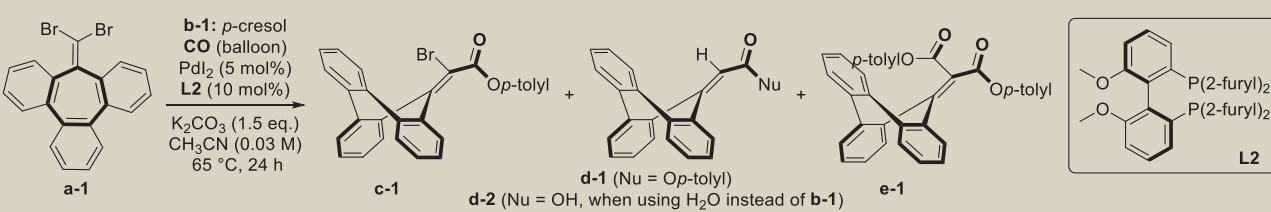
the *gem*-dibromide moiety also underwent monoesterification successfully (**c-25** and **c-26**). Notably, a furan-bridged system afforded a good yield but significantly reduced enantioselectivity (**c-27**). Remarkably, when the middle phenyl was replaced by a diphenyl alkenyl moiety, a developed inherently chiral dibenzo[7]annulene structure was obtained with moderate yield and excellent enantiocontrol (**c-28**, 97:3 er). Unsymmetrically substituted racemic substrates (**c-29** to **c-32**) were also examined. These substrates reacted efficiently, generally affording 1:1 diastereomeric mixtures with good enantioselectivities, demonstrating that pre-existing chiral elements in the substrates do not interfere with the stereodetermining process.

Given the broad applications of polyaryl-substituted tribenzo[*a,c,e*][7]annulenes in materials science<sup>41–47</sup>, several symmetric polysubstituted tribenzo[*a,c,e*][7]annulene *gem*-dibromide substrates were also explored. Due to poor solubility of the starting materials, slightly elevated temperatures were required. To prevent hydrolysis of the ester products, benzylamine was added in a one-pot procedure to generate the corresponding amides in situ. Using this method, a series of polysubstituted tribenzo[*a,c,e*][7]annulene amides (**c-33** to **c-39**) were obtained in 53%–77% yields with er values of 93:7–96:4. The

absolute configuration of **c-33** was determined by single-crystal X-ray diffraction analysis, revealing a well-defined saddle-shaped conformation.

#### Substrate scope of the double debromination reaction pathway

Next, we explored the generality of the double debromination reaction pathway (Fig. 3). For tribenzo[*a,c,e*][7]annulenes bearing diverse substituents at the 6,12- or 7,11-positions (**d-2** to **d-7**), the corresponding carboxylic acid products were obtained in 78%–97% yields, with good enantioselectivity. Substrates with sterically demanding *p*-tolyl groups reacted slowly, but they did so without compromising enantioselectivity (**d-8**). Reactions proceeded smoothly when fused aromatic or heteroaromatic rings were positioned opposite the *gem*-dibromoalkene moiety (**d-9** to **d-10**). However, for the analog featuring a non-cyclic diphenyl substituent, **d-11** was obtained in excellent yield but with poor enantioselectivity, presumably due to racemization during the reaction as a result of a low barrier for conformational inversion ( $\Delta G = 27.9$  kcal/mol). Even at reduced temperatures, only moderate enantioselectivity was achieved. Polysubstituted

**Table 1 | Optimization of Reaction Conditions<sup>a</sup>**


| entry                   | [Pd]                       | solvent                 | base                                    | yield(%) / er (c-1) | yield(%) / er (d-1 or d-2) | yield(%) (e-1) |
|-------------------------|----------------------------|-------------------------|-----------------------------------------|---------------------|----------------------------|----------------|
| 1                       | PdI <sub>2</sub>           | toluene                 | K <sub>2</sub> CO <sub>3</sub>          | trace/96.5:3.5      | /                          | /              |
| 2                       | PdI <sub>2</sub>           | THF                     | K <sub>2</sub> CO <sub>3</sub>          | trace/94:6          | 5/57.5:42.5                | /              |
| 3 <sup>b</sup>          | PdI <sub>2</sub>           | CH <sub>3</sub> CN      | K <sub>2</sub> CO <sub>3</sub>          | 20/94:6             | 9/51:49                    | 25             |
| 4                       | PdI <sub>2</sub>           | CH <sub>3</sub> CN      | NaH                                     | 15/97:2             | 11/50:50                   | /              |
| 5                       | PdI <sub>2</sub>           | CH <sub>3</sub> CN      | KOH                                     | 32/97:2             | 9/53.5:46.5                | trace          |
| 6                       | PdI <sub>2</sub>           | CH <sub>3</sub> CN      | CS <sub>2</sub> CO <sub>3</sub>         | 40/97:3             | 9/51:49                    | trace          |
| 7 <sup>c</sup>          | PdI <sub>2</sub>           | CH <sub>3</sub> CN      | CS <sub>2</sub> CO <sub>3</sub>         | 16/94:6             | 13/51.5:48.5               | 6              |
| 8                       | Pd(dba) <sub>2</sub>       | CH <sub>3</sub> CN      | CS <sub>2</sub> CO <sub>3</sub>         | /                   | /                          | /              |
| 9                       | PdCl <sub>2</sub>          | CH <sub>3</sub> CN      | CS <sub>2</sub> CO <sub>3</sub>         | 43/95:5             | 5/51:49                    | trace          |
| 10                      | Pd(OAc) <sub>2</sub>       | CH <sub>3</sub> CN      | CS <sub>2</sub> CO <sub>3</sub>         | 45/95:5             | 6/53:47                    | /              |
| 11                      | Pd(TFA) <sub>2</sub>       | CH <sub>3</sub> CN      | CS <sub>2</sub> CO <sub>3</sub>         | 27/96:4             | 5/51:49                    | /              |
| <b>12<sup>d</sup></b>   | <b>Pd(OAc)<sub>2</sub></b> | <b>CH<sub>3</sub>CN</b> | <b>CS<sub>2</sub>CO<sub>3</sub></b>     | <b>73/96.5:3.5</b>  | <b>6/50.5:49.5</b>         | <b>/</b>       |
| 13 <sup>e</sup>         | Pd(OAc) <sub>2</sub>       | CH <sub>3</sub> CN      | K <sub>2</sub> CO <sub>3</sub> +PivOH   | trace //            | 47/92:8                    | /              |
| 14 <sup>e</sup>         | Pd(OAc) <sub>2</sub>       | CH <sub>3</sub> CN      | K <sub>2</sub> CO <sub>3</sub> +AcOH    | trace //            | 35/93:7                    | /              |
| 15 <sup>e</sup>         | Pd(OAc) <sub>2</sub>       | CH <sub>3</sub> CN      | PivOH                                   | /                   | /                          | /              |
| 16 <sup>e</sup>         | Pd(OAc) <sub>2</sub>       | CH <sub>3</sub> CN      | KOPiv                                   | trace //            | 31/92:8                    | /              |
| 17 <sup>e,f</sup>       | Pd(OAc) <sub>2</sub>       | CH <sub>3</sub> CN      | KOPiv+5 Å                               | trace //            | /                          | /              |
| 18 <sup>e,g</sup>       | Pd(OAc) <sub>2</sub>       | CH <sub>3</sub> CN      | K <sub>2</sub> CO <sub>3</sub> +AcOH    | /                   | 75/91.5:8.5                | /              |
| 19 <sup>e,g</sup>       | Pd(OAc) <sub>2</sub>       | THF                     | K <sub>2</sub> CO <sub>3</sub> +AcOH    | /                   | 89/94:6                    | /              |
| <b>20<sup>e,h</sup></b> | <b>Pd(OAc)<sub>2</sub></b> | <b>THF</b>              | <b>K<sub>2</sub>CO<sub>3</sub>+AcOH</b> | <b>/</b>            | <b>93/94.5:5.5</b>         | <b>/</b>       |

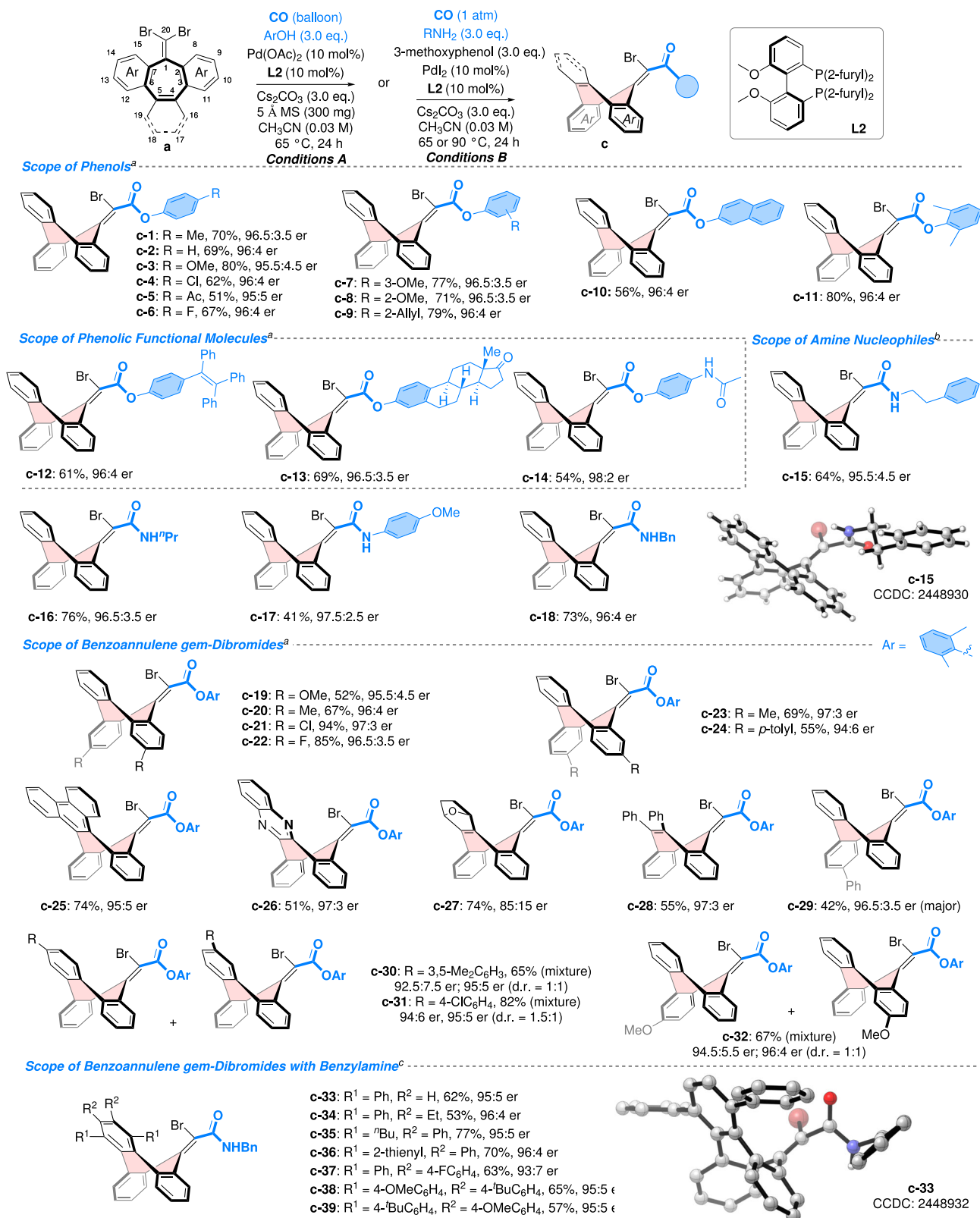
<sup>a</sup> Conditions: A flame-dried 10 mL Schlenk tube was charged with [Pd] (5 mol%), L (10 mol%), base (1.5 eq.), and a-1 (21.0 mg, 0.05 mmol). The mixture was evacuated and backfilled with CO (3 cycles). Then, b-1 (1.2 eq.) and 1.5 mL of dry CH<sub>3</sub>CN were added to the mixture via syringe. Finally, the mixture was stirred at 65 °C for 24 h under a CO balloon atmosphere. The enantiomeric ratio (er) was determined by HPLC, and the yield was determined by <sup>1</sup>H NMR spectroscopy using 1,3,5-trimethoxybenzene as an internal standard. <sup>b</sup> 48 h. <sup>c</sup> 80 °C. <sup>d</sup> [Pd]/L (10 mol%), b-1/base (3.0 eq.), 5 Å MS (150.0 mg). <sup>e</sup> Pd(OAc)<sub>2</sub>/L (10 mol%), base (1.5 eq.), additive (2.0 eq.) and b-1 (3.0 eq.). <sup>f</sup> 5 Å MS (150.0 mg). <sup>g</sup> H<sub>2</sub>O (3.0 eq.) was used instead of b-1. <sup>h</sup> H<sub>2</sub>O (1.5 eq.), 60 °C. <sup>h</sup> A sealed tube with a Teflon screw cap was used under atmospheric pressure CO, and the product is d-2.

tribenzo[*a,c,e*][7]annulene *gem*-dibromides performed poorly (d-12), likely due to their poor solubility. For racemic substrates with asymmetric substitution patterns (d-13 to d-15), the corresponding products were successfully obtained through parallel kinetic resolution, indicating that pre-existing chiral elements in the substrates exert negligible influence on the stereochemical outcome.

### Mechanistic studies on the stereoselectivity

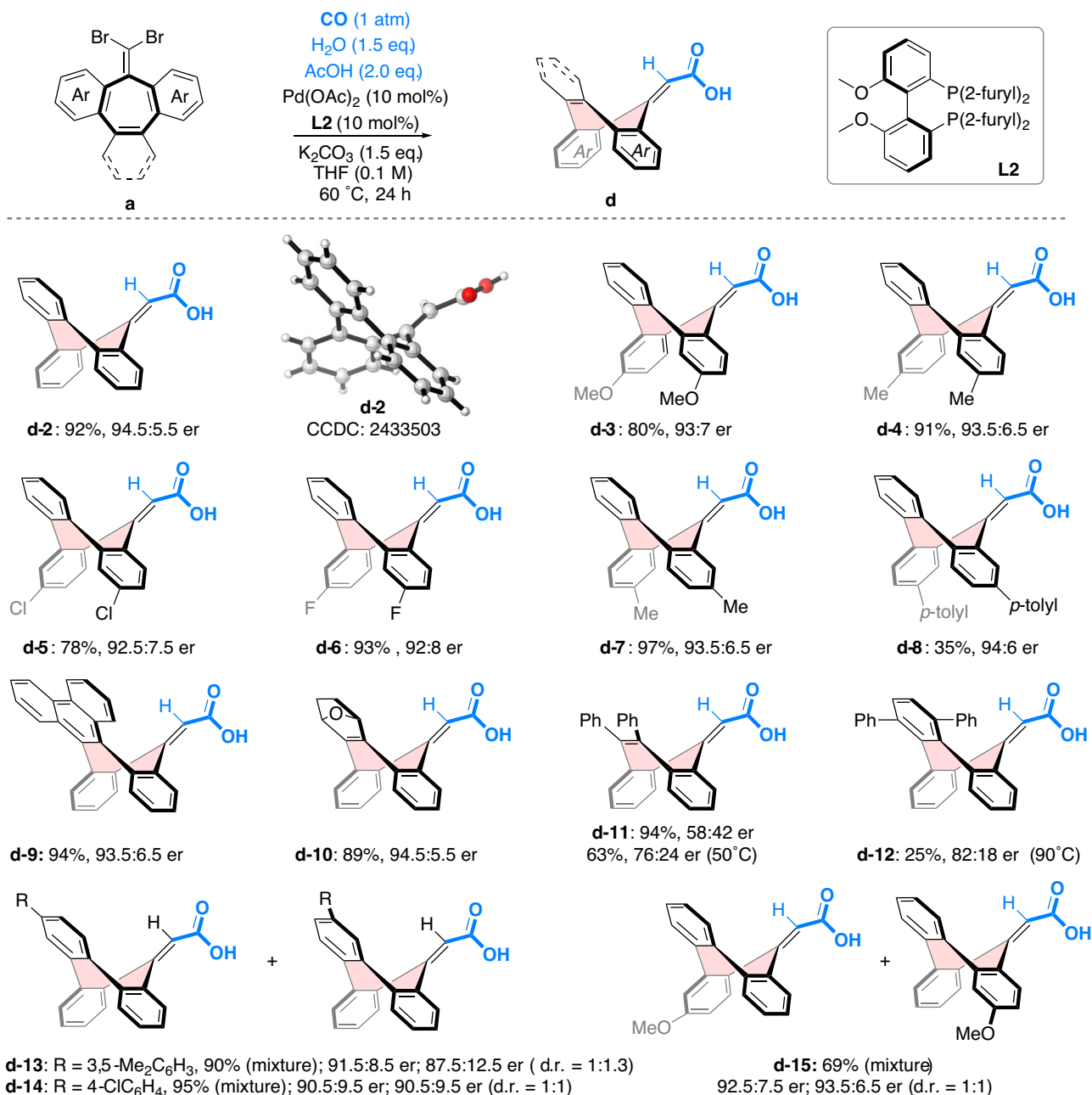
To elucidate the mechanism of these chemo- and enantioselective carbonylation reactions, we designed and conducted a series of control experiments (Fig. 4). First, nonlinear effect studies in model reactions of distinct pathways revealed a clear positive linear correlation between the enantiopurity of chiral ligand L2 and the enantioselectivity of the product (Fig. 4A). This relationship confirms the presence of mononuclear palladium catalysts coordinated with a single chiral ligand molecule indicating that a dominant catalytic species controls the asymmetric induction during carbonylation. For the monocarbonylation pathway, the mechanism appears to follow a classical reaction route (Fig. 5, Path 1). In contrast, control experiments indicate that the double debrominative carbonylation product is not formed via initial monocarbonylation to c-1 followed by

debrominative hydrogenation (Fig. 4B, Eq. 1). Further control experiments (Eqs. 3 and 4) demonstrated that the monobromide starting material exhibits significantly lower reactivity than a-1, thereby excluding a pathway involving a-1-Br as the reaction intermediate. Therefore, we propose that after oxidative addition of the *gem*-dibromoalkene to palladium to generate INT-1, competitive ligand exchange and α-Br elimination with a trace amount of Pd-H species present in the system occur, affording the palladium carbene intermediate INT-5 (Fig. 5, Path 1). This step was directly validated by DFT calculations in Path 2, and is also supported by literature precedents for the formation of related alkylidene carbene species<sup>48–52</sup>. Subsequent CO migratory insertion then forms an alkylidene ketene intermediate INT-6, which was detected by HRMS. Attack of this intermediate by the phenol nucleophile present in the system ultimately delivers the debrominated product d-1<sup>53,54</sup>. The formation of d-1 from the reactive intermediate INT-6 was a non-enantioselective process, which was supported by the fact that d-1 was always obtained in nearly racemic forms as observed in Table 1. Finally, DFT calculations were conducted at the PWPB95-D3(BJ)/def2-TZVPP (SMD, MeCN)//PBE0-D3(BJ)/6-31G(d)/SDD level of theory, and the energy profile of the monocarbonylation pathway was established as illustrated in Fig. 6. The oxidative addition of the C-Br bond in



**Fig. 2 | Substrate scope of the stereoselective mono-carbonylation of Benzo[7]annulene gem-Dibromides.** <sup>a</sup> Conditions A: **a** (0.1 mmol), phenols (0.3 mmol), Pd(OAc)<sub>2</sub> (10 mol%), **L2** (10 mol%), Cs<sub>2</sub>CO<sub>3</sub> (0.3 mmol), 5 Å MS (300 mg), dry MeCN (3.0 mL), CO (balloon), 65 °C, 24 h; <sup>b</sup> Conditions B: **a** (0.1 mmol),

amines (0.3 mmol), *m*-methoxyphenol (0.3 mmol), PdI<sub>2</sub> (10 mol%), **L2** (10 mol%), Cs<sub>2</sub>CO<sub>3</sub> (0.3 mmol), dry MeCN (3.0 mL), CO, 90 °C, 24 h; <sup>c</sup> Conditions C: **a** (0.1 mmol), amines (0.3 mmol), *m*-methoxyphenol (0.3 mmol), PdI<sub>2</sub> (10 mol%), **L2** (10 mol%), Cs<sub>2</sub>CO<sub>3</sub> (0.3 mmol), dry MeCN (3.0 mL), CO, 90 °C, 24 h; isolated yield based on **a**; er and diastereomeric ratio (dr) were determined by HPLC analysis.



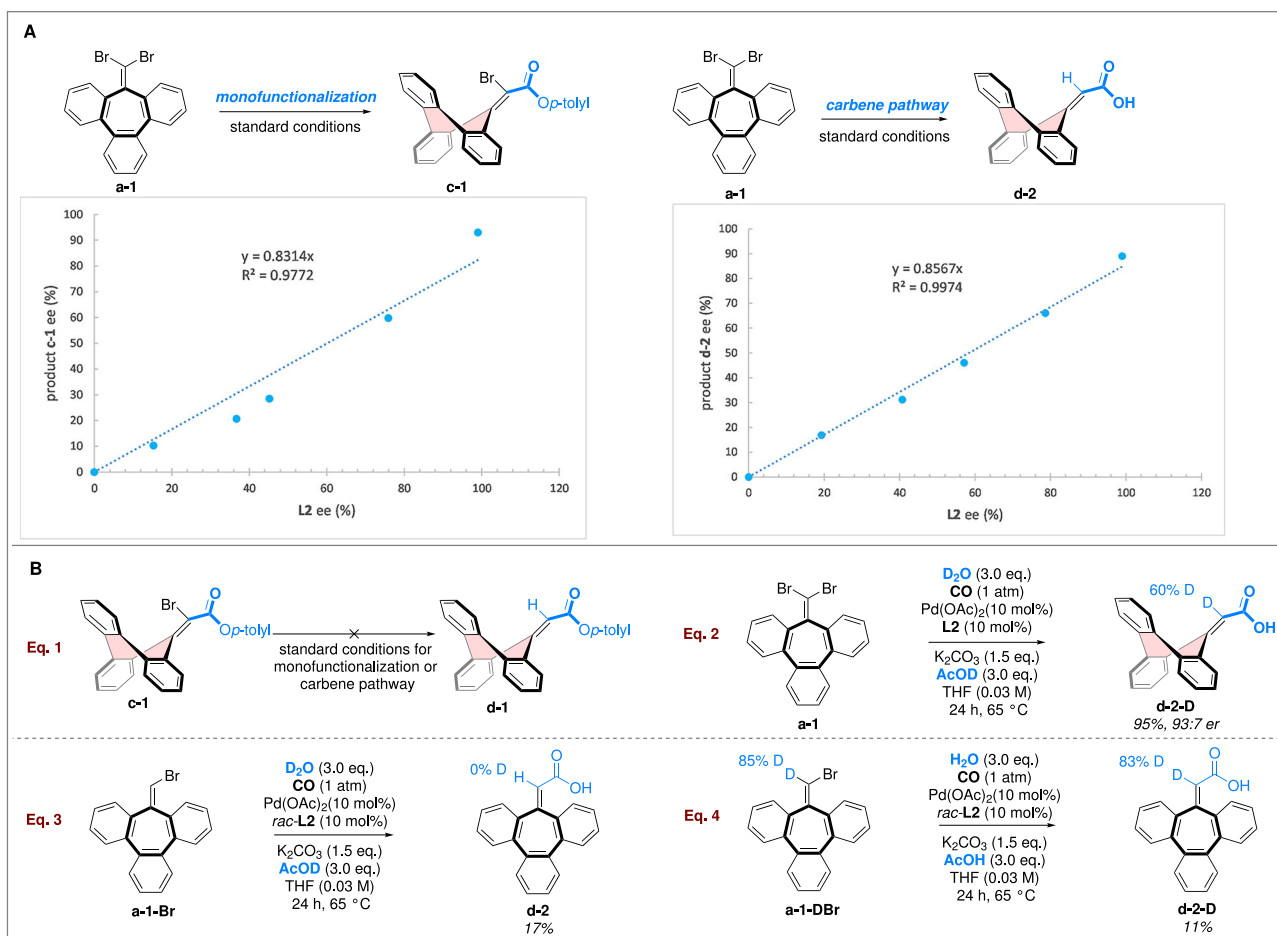
**Fig. 3 | Substrate scope of the debromination pathway of benzo[7]annulene gem-dibromides.** Conditions: **a** (0.1 mmol),  $\text{H}_2\text{O}$  (0.15 mmol),  $\text{Pd}(\text{OAc})_2$  (10 mol%),

**L2** (10 mol%),  $\text{AcOH}$  (0.2 mmol),  $\text{K}_2\text{CO}_3$  (0.15 mmol), dry THF (1.0 mL),  $\text{CO}$ ,  $60^\circ\text{C}$ , 24 h; isolated yield based on **a**; er and dr were determined by HPLC analysis.

substrate **a-1** by  $\text{Pd}(0)$  species is identified as the enantio-determining step of the entire process. The energy difference between the two transition states (**TS-1-Major** vs **TS-1-Minor**) is 1.5 kcal/mol, which aligns well with the experimental results. Distortion-interaction analysis of these transition states reveals that the major contribution of the activation energy ( $\Delta\Delta E_{\text{act}}$ ) stems from the interaction energies ( $\Delta\Delta E_{\text{int}}$ ) between the palladium catalyst fragment and substrate fragment. Structural analysis of these two transition states reveals that for **TS-1-Major**, the C–Br bond is already elongated while the C–Pd bond is in the process of formation, rendering the overall transition state structure more product-like (resembling intermediate **INT-1**); in contrast, for **TS-1-Minor**, the C–Br bond retains its initial length, and the C–Pd bond distance remains considerable, indicating that this transition state is more reactant-like in character. Subsequent

to the formation of the oxidative addition intermediate **INT-1**, the reaction proceeds through CO coordination, migratory insertion, ligand exchange, and reductive elimination, ultimately furnishing the target product.

On the contrary, structurally similar double-debrominated carbonylation product **d-2** was obtained in good enantioselectivity in the presence of acidic additives, suggesting that acid altered the reaction pathway, avoiding the aforementioned alkylidene ketene intermediate. Therefore, we propose the following mechanism by combining controlled experiments and DFT calculations (Fig. 7). First, **INT-1** undergoes ligand exchange with **HPdLOH**, which is generated via the oxidative addition of  $\text{AcOH}$  to  $\text{Pd}(0)$  followed by a subsequent ligand exchange step<sup>35</sup>, to afford the  $\text{Pd-H}$  species **INT-1'**. Next, **INT-1'** undergoes  $\alpha$ -Br elimination to generate the palladium carbene intermediate **INT-5** via **TS-4** (the carbene homocoupling byproduct was



**Fig. 4 | Mechanistic studies. A** Non-linear effect experiments. **B** Control Experiments. Isolated yield; er was determined by HPLC analysis.

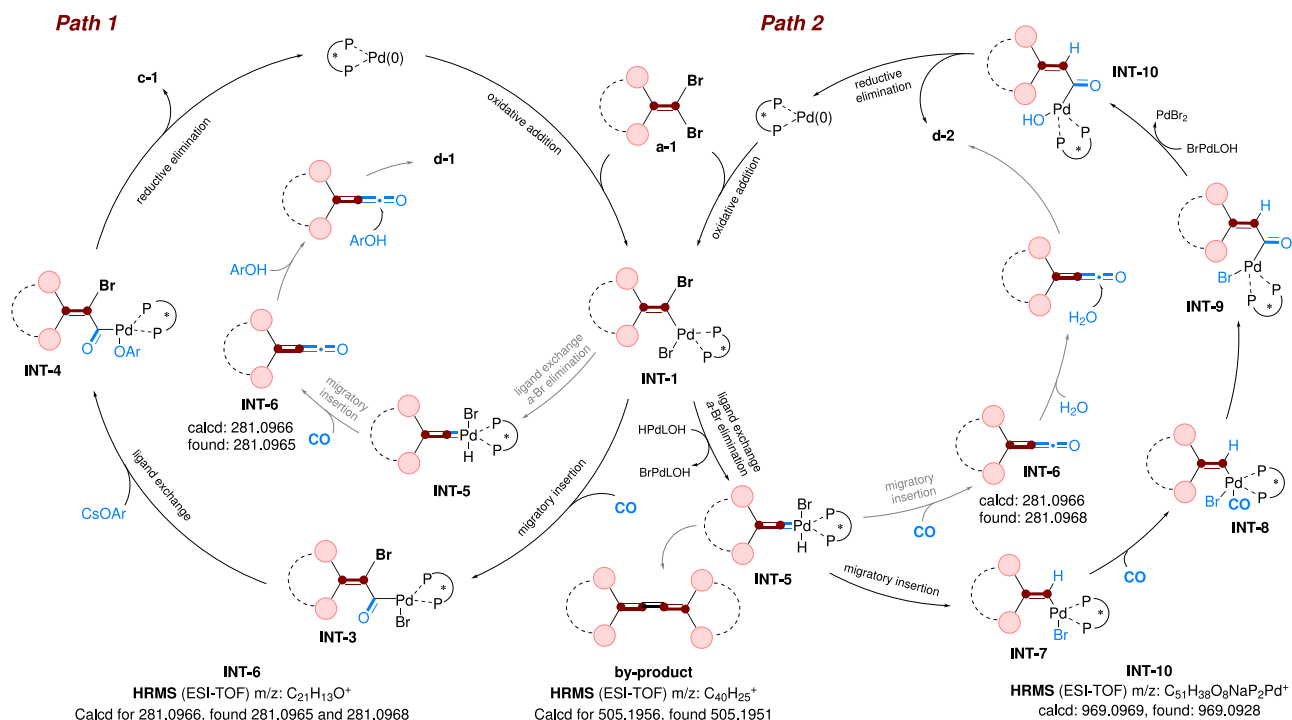
detected by HRMS). The activation energy barrier for this step is 26.7 kcal/mol, making it kinetically favored over the CO migratory insertion pathway described in Path 1. Next, the Pd-carbene undergoes migratory insertion into the H–Pd bond via a three-membered ring transition state (**TS-5**), forming the alkenyl palladium intermediate (**INT-7**). This carbene insertion is the enantio-determining step, and the computed energy difference between the two transition states (**TS-5-Major** vs **TS-5-Minor**) is 2.1 kcal/mol, which also aligns well with the experimental results. Distortion-interaction analysis of these transition states reveals that the major contribution of the activation energy ( $\Delta\Delta E_{act}$ ) stems from the distortion of the palladium catalyst fragment ( $\Delta\Delta E_{Pd}$ ). Notably, this migratory insertion step is highly exergonic, with the generated **INT-7** having an energy as low as -47.8 kcal/mol. Subsequently, through CO migratory insertion, ligand exchange, and reductive elimination (**INT-10** was detected by HRMS), the double debrominative carbonylation product is ultimately obtained. However, we cannot rule out the possibility that some portion of the product may still be generated via the alkylidene ketene intermediate **INT-6** (Fig. 5, Path 2, detected by HRMS, and the relative free energy is -10.6 kcal/mol), which could explain why the enantioselectivity of **d-2** does not reach an excellent level.

The regulatory effect of acidic additives on divergent reaction pathways likely stems from different rates of CO migratory insertion, the nucleophilicity of phenol, and the generation of Pd–H species. Under basic conditions, phenolic substrates exist as highly nucleophilic phenoxide anions which facilitates ligand exchange with bromide followed by rapid CO migratory insertion to suppress the  $\alpha$ -Br elimination process. In contrast, under weakly basic or acidic conditions, phenolic substrates are less nucleophilic, which slows CO insertion.

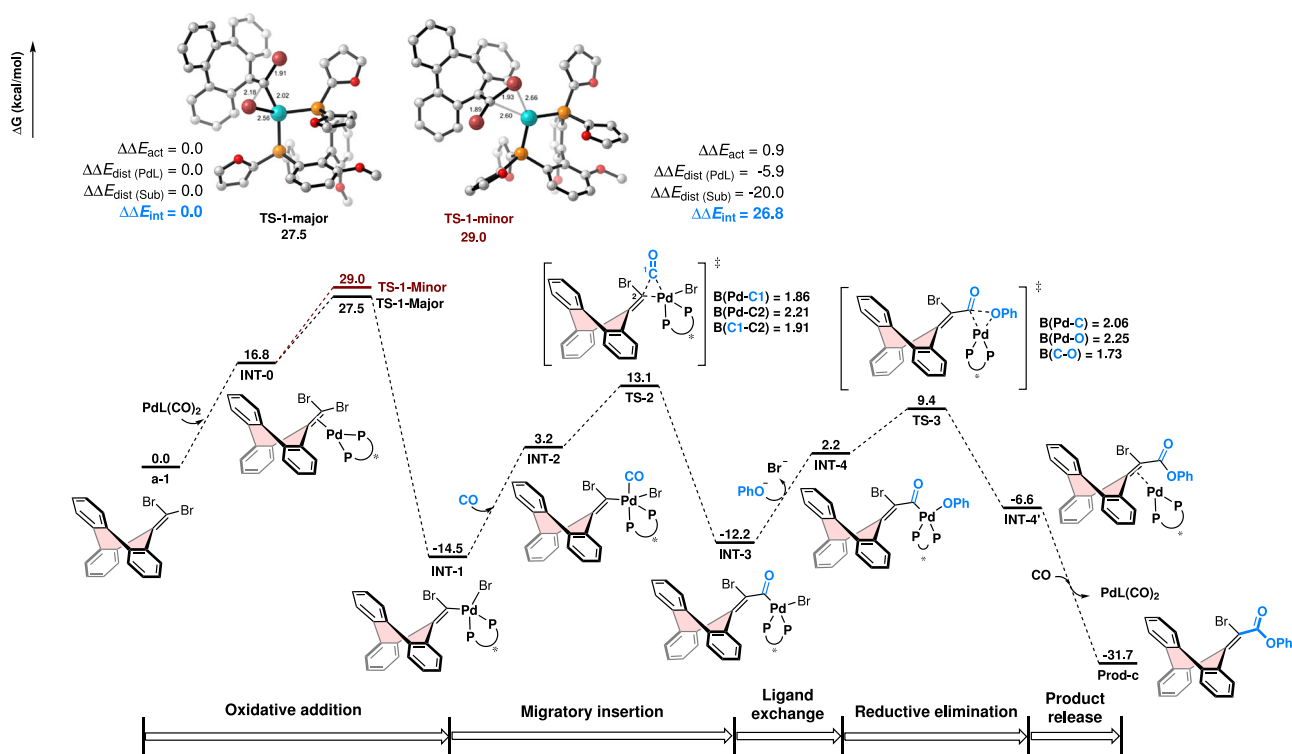
The delay allows  $\alpha$ -Br elimination to form vinylpalladium carbene intermediates, directing the reaction toward Path 2. At the same time, the acidic environment and the presence of CO facilitate the generation of palladium hydride species. This mechanism differs from the Ni-catalyzed reductive vinyl carbene transfer reaction reported by the Uyeda group<sup>56–63</sup>. Deuterium labeling experiments provided additional evidence: when monobromoalkene was used as the substrate, the reaction did not proceed through the vinyl carbene pathway, as the vinylic hydrogen pattern in the product remained identical to that of the starting material (Fig. 4B, Eqs. 3 and 4)<sup>64</sup>.

### Synthetic utility

Dibenzofulvene derivatives have attracted considerable attention in chemistry and materials science owing to their distinctive optoelectronic properties<sup>65</sup>. Starting from product **c-31**, hydrolysis afforded  $\alpha$ -brominated tribenzo[*a,c,e*][7]annulene carboxylic acid **c-31-COOH**. Employing a known reaction, a structurally unique inherently chiral tribenzo[*a,c,e*][7]annulene-based dibenzofulvene **c-31-fluorene** was successfully constructed in a single step with complete retention of enantiopurity, and its absolute configuration was determined by single-crystal X-ray diffraction analysis (Fig. 8, top). The monocarbonylation reaction was successfully scaled up to a gram scale, delivering 69% yield and 96.5:3.5 er. Hydrolysis of **c-1** gave  $\alpha$ -brominated tribenzo[*a,c,e*][7]annulene carboxylic acid **c-1-COOH** in 88% yield, with its absolute configuration determined by single-crystal X-ray diffraction (Fig. 8). Using **c-1-COOH** as a key synthon, diverse tetrasubstituted inherently chiral tribenzo[*a,c,e*][7]annulene-based olefins were synthesized (See Supplementary Information, Synthetic Transformations section, for details). For example, a one-pot Suzuki



**Fig. 5 | Catalytic cycle. Path 1:** catalytic cycle of monocarbonylation pathway. **Path 2:** catalytic cycle of double debrominative carbonylation.

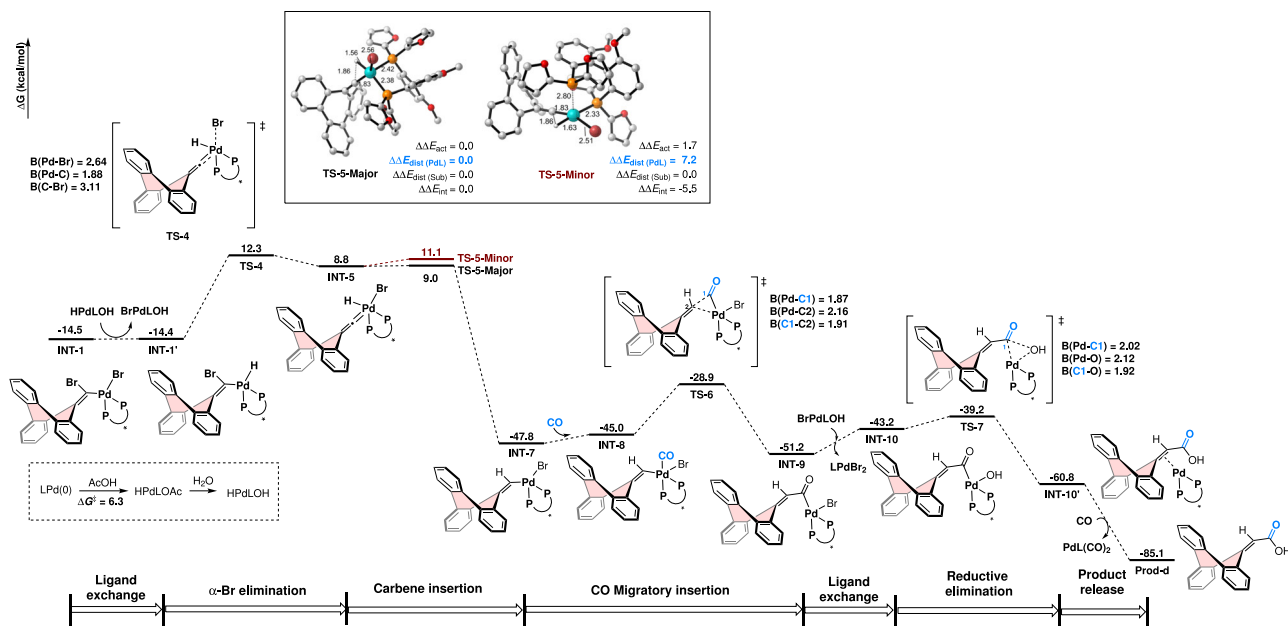


**Fig. 6 | Energy profiles of the Pd-catalyzed monocarbonylation pathway (Path 1).** Calculations performed at the PWPB95-D3(BJ)/def2-TZVP (SMD, MeCN)//PBE0-

D3(BJ)/6-31 G(d)/SDD level of theory. Distances are in angstroms.

coupling followed by in situ amidation yielded **g-1**, which incorporates an indolin-2-one moiety. Quantitative methylation of the carboxyl group yielded **f-1**, and subsequent reduction with DIBAL-H produced the inherently chiral tribenzo[*a,c,e*][7]annulene propanol **f-2**. In addition, a series of inherently chiral triaryl olefin carboxylic acids **i-1** to **i-12** was obtained via Suzuki coupling of the vinyl bromide moiety.

Subsequent decarboxylative coupling yielded optically active, inherently chiral molecule **i-13**, albeit with slightly reduced enantioselectivity (93.5:6.5 er). Amidation with a phosphinoamine fragment bearing catalytic activity afforded **i-14**. Finally, investigations were conducted to explore whether this  $\alpha$ -bromo carboxylic acid synthon could be elaborated into inherently chiral oxazoline-containing



**Fig. 7 | Energy profiles of the Pd-catalyzed double debrominative carbonylation pathway (Path 2).** Calculations performed at the PWPB95-D3(BJ)/def2-TZVPP

(SMD, MeCN)//PBE0-D3(BJ)/6-31 G(d)/SDD level of theory. Distances are in angstroms.

ligands. Condensation with a  $\beta$ -amino alcohol fragment afforded **h-1**, which was recrystallized to 99.5:0.5 er and cyclized to  $\alpha$ -bromo oxazoline **h-2**. A Stille coupling constructed **h-3**, a potential *N*-alkenyl ligand. Subsequently, Suzuki coupling followed by diphenylphosphine substitution delivered a designed inherently chiral *N,P*-ligand **h-4** with full retention of optical purity. Notably, all these inherently chiral tribenzo[*a,c,e*][7]annulene carboxylic acids can function as chiral carboxylic acid ligands, cooperating with Ru or Ir complexes to catalyze C–H activation/cyclization reactions while exhibiting moderate enantioselectivity (Fig. 8, Reactions A and B). In addition, we also evaluated the performance of **h-3** (*P*-olefin-ligand), as well as **h-4** and **i-14** (*P,N*-ligands), in asymmetric catalysis. Specifically, we preliminarily assessed these ligands in Rhodium-catalyzed asymmetric conjugate addition reactions and asymmetric allylic substitution reactions. Unfortunately, the enantioselectivities obtained were generally low (76% yield, 55.5:44.5 er) to moderate (95% yield, 77:23 er) (See Supplementary Information, Synthetic Transformations section). Although these preliminary results were not satisfactory, we believe they provide useful reference information for future optimization of these scaffolds.

## Discussion

In summary, we have developed an efficient method for the chemo- and enantioselective carbonylation of benzo[7]annulene-based *gem*-dibromoalkenes using a chiral palladium catalyst, enabling the construction of inherently chiral olefin frameworks with stereodefined alkene geometry in excellent yields (up to 97%) and high enantioselectivities (up to 98:2 er). Notably, by simply modulating acid additives, the reaction pathway can be precisely switched to double debrominative carbonylation, thereby providing divergent access to inherently chiral trisubstituted vinyl carboxylic acids with equally good chemo- and enantiocontrol. This protocol exhibits excellent substrate generality for various phenolic nucleophiles and structurally diverse benzo[7]annulene derivatives. Mechanistic studies, supported by DFT calculations, reveal that the acid-induced formation of a key alkylidene carbene–Pd–H species is pivotal for the pathway switch, followed by enantioselective carbene insertion and CO insertion. The obtained chiral products can be further diversified into functional

tetrasubstituted alkene molecules; preliminary evaluations as chiral ligands in asymmetric catalysis have shown promising yet moderate results (up to 71% yield, 81:19 er), indicating their potential for further optimization. This work not only provides a modular and divergent route to both tri- and tetrasubstituted inherently chiral benzo[7]annulene scaffolds, but also represents the first example of asymmetric desymmetrization of prochiral *gem*-dibromoalkenes via palladium catalysis. Ongoing efforts in our laboratory are directed toward further mechanistic elucidation and exploration of synthetic applications.

## Methods

### Catalytic enantioselective synthesis of **c**

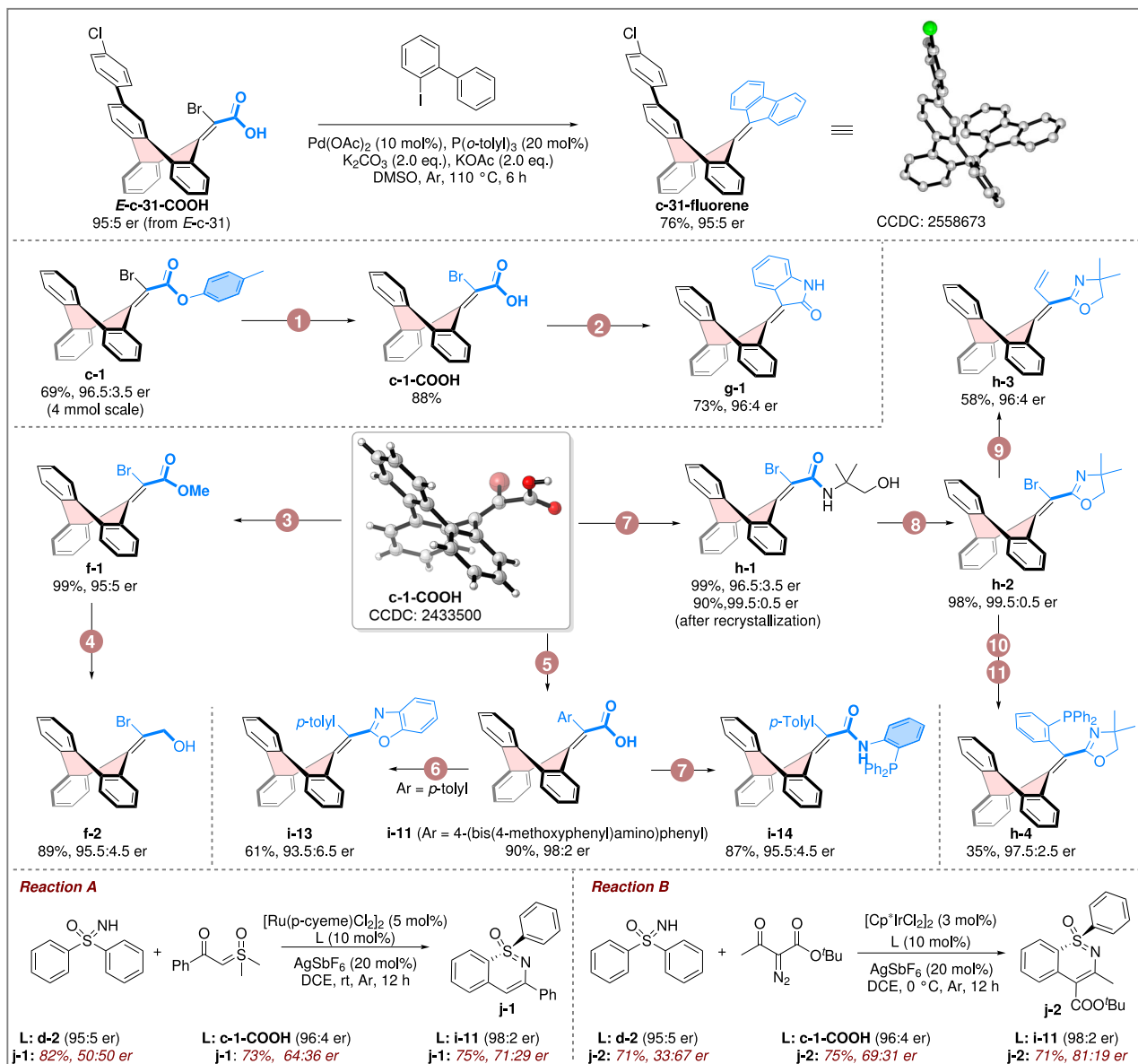
In a dried 10 mL Schlenk tube were combined  $Pd(OAc)_2$  (10 mol%), **L2** (10 mol%),  $Cs_2CO_3$  (98.0 mg, 3.0 eq.), **a** (0.1 mmol), and 5 Å MS (300 mg). The mixture was evacuated and backfilled with CO (three cycles). Then, **b** (0.3 mmol, 3.0 eq.) and 3.0 mL of dry  $CH_3CN$  were added via syringe. Finally, the reaction mixture was stirred at 65 °C for 24 h under a CO balloon. The reaction mixture was concentrated and purified by silica gel column chromatography to afford product **c**. The er of the product was analyzed by HPLC.

### Catalytic enantioselective synthesis of **d**

In a sealed tube with a Teflon screw cap were combined  $Pd(OAc)_2$  (10 mol%), **L2** (10 mol%),  $K_2CO_3$  (21.0 mg, 1.5 eq.), and **a** (0.1 mmol). The mixture was evacuated and backfilled with CO (three cycles). Then,  $AcOH$  (0.2 mmol, 2.0 eq.),  $H_2O$  (0.15 mmol, 1.5 eq.), and 1.0 mL of dry THF were added via syringe. Finally, the reaction mixture was stirred at 60 °C for 24 h under atmospheric pressure CO. The reaction mixture was concentrated and purified by silica gel column chromatography (DCM/ $CH_3OH$ ) to afford product **d**. The er of the product was analyzed by HPLC.

## Data availability

The authors declare that the data supporting the findings of this study are available within the article and Supplementary Information file. The X-ray crystallographic coordinates for structures reported in this study have been deposited at the Cambridge Crystallographic Data Center



**Fig. 8 | Scale-up synthesis and product transformation of *gem*-dibromoalkene monocarbonylation.** Conditions: 1: 1) KOH, EtOH, 50–55 °C, 40 min; 2) HCl, 1.5 h. 2: 2-aminophenylboronic acid pinacol ester (1.5 eq.), Pd(PPh<sub>3</sub>)<sub>4</sub> (5 mol%), Na<sub>2</sub>CO<sub>3</sub> (2.5 eq.), dioxane/H<sub>2</sub>O, Ar, 110 °C, 24 h. 3: TMSCHN<sub>2</sub> (3.0 eq.), Et<sub>2</sub>O/CH<sub>3</sub>OH, rt, 10 min. 4: DIBAL-H, DCM, -78 °C, 6 h. 5: (4-(bis(4-methoxyphenyl)amino)phenyl)boronic acid (1.2 eq.), Pd(PPh<sub>3</sub>)<sub>4</sub> (5 mol%), Na<sub>2</sub>CO<sub>3</sub> (2.5 eq.), dioxane/H<sub>2</sub>O, Ar, 85 °C, 12 h. 6: benzo[d]oxazole (1.0 eq.), Pd(OAc)<sub>2</sub> (4 mol%), CuCl (10 mol%), DPPP (10 mol%),

Boc<sub>2</sub>O (1.5 eq.), NaHCO<sub>3</sub> (1.5 eq.), cyclohexane, 120 °C, 12 h. 7: 1) (COCl)<sub>2</sub>, DMF; 2) 2-amino-2-methylpropan-1-ol or 2-(diphenylphosphanyl)aniline (1.5 eq.), Et<sub>3</sub>N. 8: PPh<sub>3</sub>, Et<sub>3</sub>N, CCl<sub>4</sub>, MeCN, rt, 12 h. 9: tributyl(vinyl)tin (1.3 eq.), Pd-PEPPSI-IPr (5 mol%), CsF (2.0 eq.), 5 Å MS, THF, 80 °C, 12 h. 10: 2-(2-bromophenyl)-4,4,5,5-tetramethyl-1,3,2-dioxaborolane (1.3 eq.), Pd(PPh<sub>3</sub>)<sub>4</sub> (10 mol%), Na<sub>2</sub>CO<sub>3</sub> (2.5 eq.), dioxane/H<sub>2</sub>O, Ar, 110 °C, 6 h. 11: 1) *n*-BuLi, THF, -78 °C, 1 h; 2) PPh<sub>2</sub>Cl (1.2 eq.), 12 h. Isolated yield; er was determined by HPLC analysis.

(CCDC), under deposition numbers CCDC number 2448930 (**c-15**), 2448932 (**c-33**), 2433503 (**d-2**), 2433500 (**c-1-COOH**) and 2558673 (**c-31-fluorene**). Data supporting the findings of this manuscript are also available from the corresponding author upon request.

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## Author contributions

Y.L. performed most of the experimental studies. X.W. and Y.Z. conducted the MS spectrum experiments. S.L. provided support for the DFT calculations. Q.Z. directed the project and wrote the paper with the feedback from other authors.

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## Competing interests

The authors declare no competing interests.

## Additional information

**Supplementary information** The online version contains supplementary material available at (<https://doi.org/10.1038/s41467-026-78226-0>).

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